

ASYN4 Machine Performance Report

Field	Value
NAME	TEST-MACHINE
REKE	M-TYPE-FRAME
DATUM	10072026

Partial-Load Performance

Load	P_mech (kW)	Eta	cos(phi)	I1 (A)	Slip (%)	Pcu1 (kW)	Pcu2 (kW)	M (Nm)
1/4	2.00	0.9037	0.652	2.96		0.015	0.022	12.9
2/4	0.50	0.7316	0.274	2.17		0.008	0.003	3.2
3/4	1.00	0.8411	0.441	2.36		0.009	0.006	6.4
4/4 (rated)	1.50	0.8826	0.564	2.63		0.011	0.013	9.6
5/4	2.49	0.9151	0.711	3.34		0.018	0.034	16.1

Breakdown / Locked-Rotor Summary

Quantity	Value
Breakdown slip (Kipp)	9.382 %
Breakdown torque	62.4 Nm
Locked-rotor torque (theoretical)	15.5 Nm
Locked-rotor current (theoretical)	19.9 A
Locked-rotor torque (corrected, KKA=0.79)	17.6 Nm
Locked-rotor current (corrected)	21.2 A
Time constant TK	0.0671 s
Time constant THAUPT (rotor)	0.2966 s
Time constant TSTREU (leakage)	0.0224 s

Magnetization Characteristic

Generated by the magnetic-circuit solver (KREIS) as air-gap induction is swept from below rated voltage up to 150% of rated voltage.

U _I (V)	I _{my} (A)	B _p (T)	P _{fe1} (kW)	B _{joch1} (T)	B _{joch2} (T)	B _{z1max} (T)	B _{z2max} (T)
0.1	0.00	0.000	0.000	0.000	0.000	0.000	0.000
80.6	0.45	0.076	0.008	0.182	0.182	0.157	0.283
160.6	0.89	0.151	0.031	0.363	0.363	0.314	0.566
240.6	1.32	0.226	0.070	0.543	0.543	0.472	0.849
320.8	1.78	0.302	0.124	0.725	0.725	0.629	1.132
347.6	1.95	0.327	0.146	0.785	0.785	0.681	1.226
356.2	2.00	0.335	0.153	0.805	0.805	0.698	1.256
364.9	2.06	0.343	0.160	0.824	0.824	0.715	1.286
373.6	2.11	0.352	0.168	0.844	0.844	0.731	1.316
382.3	2.16	0.360	0.176	0.864	0.864	0.748	1.347
391.1	2.22	0.368	0.184	0.883	0.883	0.765	1.377
449.0	2.66	0.422	0.242	1.014	1.014	0.870	1.565
519.6	3.47	0.489	0.321	1.174	1.174	0.974	1.754
608.3	4.97	0.572	0.434	1.374	1.374	1.079	1.942

Synchronous (Cusp) Torque Analysis

Maximum standstill synchronous torque: 0.0997 x Mn. Starting endangered: False.

Pole-pair number NY	Msy/Mn (standstill)
-22	0.0093
26	-0.0181
-46	-0.0001
50	-0.0001
-70	-0.0454
74	-0.0429
-94	-0.0000
98	-0.0001
-118	-0.0040
122	0.0017

Cooling & Material Information

Quantity	Value
Dominant cooling path (IKLINK)	0
Stator yoke iron area AR1	0.05870 m2
Stator tooth iron area AZ1	0.01068 m2
Rotor yoke iron area AR2	-0.01325 m2
Rotor tooth iron area AZ2	0.01152 m2

Quantity	Value
Stator lamination	Vollblech
Rotor lamination	Vollblech